

BALTIMORE-WASHINGTON, MD, USAWMO: **724060**Lat: **39.173N**Lon: **76.684W**Elev: **156**StdP: **14.61**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **93721****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	17.5	-3.6	4.6	17.6	1.1	5.9	20.6	26.3	35.5	24.0	33.3	8.0	290	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	94.0	75.0	91.3	74.2	88.7	72.9	78.0	88.8	76.7	86.4	75.6	84.0	9.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	132.7	81.6	74.1	127.8	80.5	73.0	123.2	79.7	41.4	89.2	40.1	86.4	39.0	84.2	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	21.9	18.8	16.6	DB	7.0	98.1	4.6	3.5	3.7	100.6	1.0	102.7	-1.5	104.7	-4.9	107.3
(5)				WB	5.4	80.2	4.4	1.8	2.3	81.5	-0.3	82.5	-2.7	83.5	-5.9	84.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.3	34.0	36.5	44.4	55.3	64.4	73.5	77.9	76.1	69.4	57.7	46.9	38.8	
	DBStd	17.23	10.14	9.30	9.40	8.49	7.75	6.22	5.02	4.89	6.53	8.03	8.21	8.61	
	HDD50	1698	505	388	228	37	1	0	0	0	0	19	155	365	
	HDD65	4475	961	797	642	311	111	9	0	0	31	255	544	814	
	CDD50	4011	10	11	53	197	446	705	865	808	581	257	62	16	
	CDD65	1314	0	0	2	21	91	265	401	343	162	28	1	0	
	CDH74	12166	0	2	32	250	859	2460	4038	3085	1226	207	7	0	
	CDH80	4619	0	0	5	70	276	936	1749	1164	375	44	0	0	
Wind	WSAvg	6.6	7.4	7.7	8.3	8.0	6.7	6.1	5.5	5.2	5.7	5.9	6.4	6.7	
Precipitation	PrecAvg	42.8	3.0	3.0	3.7	3.2	3.9	3.8	4.1	3.9	4.0	3.4	3.2	3.5	
	PrecMax	71.8	7.8	7.2	8.7	8.6	8.7	13.1	16.7	10.9	13.3	9.2	8.1	8.1	
	PrecMin	28.2	0.5	0.3	0.2	0.4	0.4	0.8	0.7	0.9	0.2	0.0	0.3	0.6	
	PrecStd	8.7	1.5	1.7	1.7	1.5	2.0	2.3	2.6	2.2	2.9	2.2	1.9	1.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	70.1	78.7	87.0	90.7	95.7	98.4	96.2	92.3	84.8	74.1	68.8	
		MCWB	58.0	57.8	61.0	66.5	72.0	74.9	76.9	76.0	72.7	70.5	62.6	61.1	
	2%	DB	60.5	62.6	71.0	81.3	86.8	91.8	94.8	92.4	88.2	79.7	69.0	62.2	
		MCWB	55.2	52.4	57.8	62.9	70.3	74.0	75.7	75.1	71.8	67.7	59.5	56.8	
	5%	DB	54.7	56.1	65.4	75.6	82.9	88.6	91.8	89.3	84.3	75.1	64.7	57.0	
		MCWB	49.0	47.7	54.3	60.9	68.5	72.7	74.8	73.7	70.4	65.4	57.2	51.5	
	10%	DB	48.7	50.7	60.2	70.8	78.9	85.5	88.8	86.3	80.8	71.4	60.8	51.9	
		MCWB	43.4	44.0	50.8	58.7	66.3	71.6	73.7	72.3	68.8	64.0	54.2	46.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	61.6	63.5	69.4	74.9	78.6	80.0	79.2	76.9	74.2	66.2	63.5
			MCDB	63.5	68.4	72.8	78.9	86.6	88.6	92.6	90.2	85.1	78.9	70.2	67.0
2%		WB	56.3	54.7	59.9	66.4	72.5	76.4	78.2	77.5	75.2	70.9	62.5	58.2	
		MCDB	60.1	60.4	68.1	76.3	83.2	86.3	89.9	88.0	82.1	76.7	66.7	61.7	
5%		WB	49.9	48.9	56.3	63.5	70.4	75.2	77.0	76.2	73.7	68.3	59.0	52.7	
		MCDB	53.9	54.3	63.7	72.7	79.9	84.5	87.4	84.8	79.9	73.6	63.2	55.8	
10%		WB	44.3	44.6	52.0	60.5	68.4	73.8	75.8	75.0	72.1	64.5	55.4	47.6	
		MCDB	47.8	50.3	58.6	69.3	76.2	82.2	84.9	82.3	77.8	69.7	59.4	51.0	
Mean Daily Temperature Range	5% DB	MDBR	15.5	17.1	18.7	20.7	19.7	19.1	18.6	18.2	18.4	19.3	18.4	15.7	
		MCDBR	22.9	25.8	26.8	27.8	24.7	22.6	21.8	21.2	22.6	23.5	23.9	21.9	
		MCWBR	17.7	17.9	16.7	14.1	11.4	8.9	7.4	7.7	9.3	12.7	15.6	17.3	
	5% WB	MCDBR	21.4	22.8	24.0	23.2	21.7	19.2	18.5	18.0	17.0	18.4	19.0	19.1	
		MCWBR	18.4	18.0	17.0	13.5	10.9	8.6	7.4	7.5	8.3	11.8	15.1	17.4	
Clear-Sky Solar Irradiance	taub	0.309	0.320	0.357	0.398	0.441	0.494	0.514	0.509	0.426	0.361	0.324	0.309		
	taud	2.477	2.430	2.391	2.298	2.207	2.099	2.062	2.061	2.266	2.451	2.515	2.514		
	Ebn at Noon	275	287	285	279	268	252	246	244	259	268	267	266		
	Edh at Noon	25	30	35	41	45	50	52	51	39	29	24	23		
All-Sky Solar Radiation	RadAvg	647	918	1218	1559	1747	1937	1913	1696	1382	1025	739	567		
	RadStd	53	105	90	124	206	148	95	95	161	112	62	55		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days		
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